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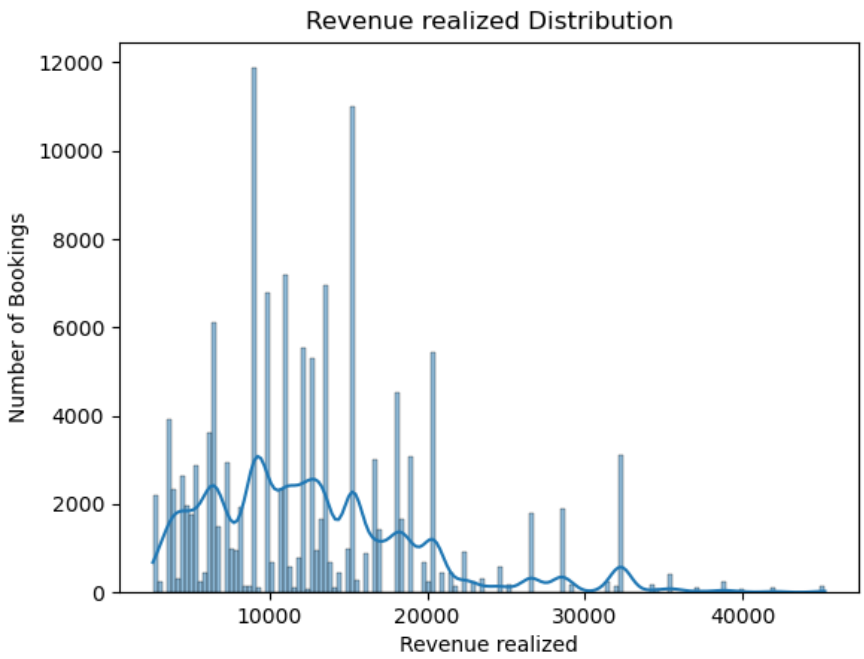
Performed Exploratory Data Analysis (EDA) on three months of hotel booking data (~135,000 records) to identify revenue trends, guest behavior patterns, and overall hotel performance. The process included data cleaning, validation, and preparation to ensure accuracy and reliability.

Analyzed revenue distribution, guest ratings, booking patterns, cancellations, and performance across cities and hotel segments to generate actionable insights. The analysis revealed key revenue drivers, seasonal trends, high- and low-performing hotels, and opportunities to optimize operations, enhance marketing strategies, and improve guest satisfaction.

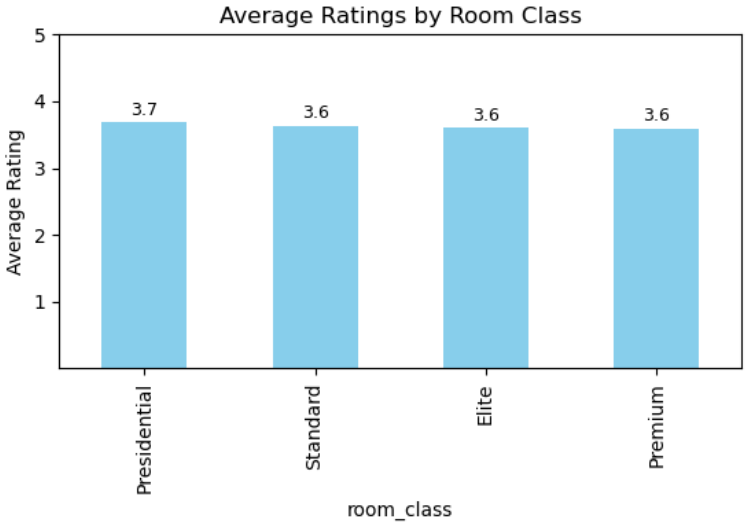
1.REVENUE DISTRIBUTION

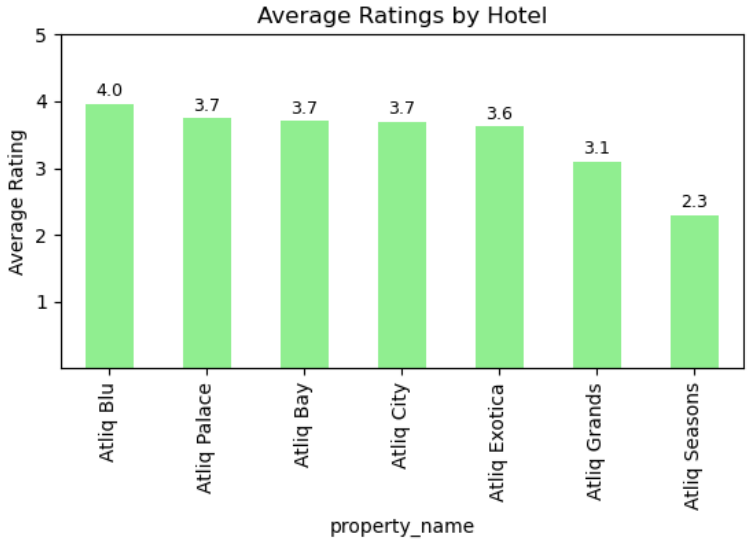


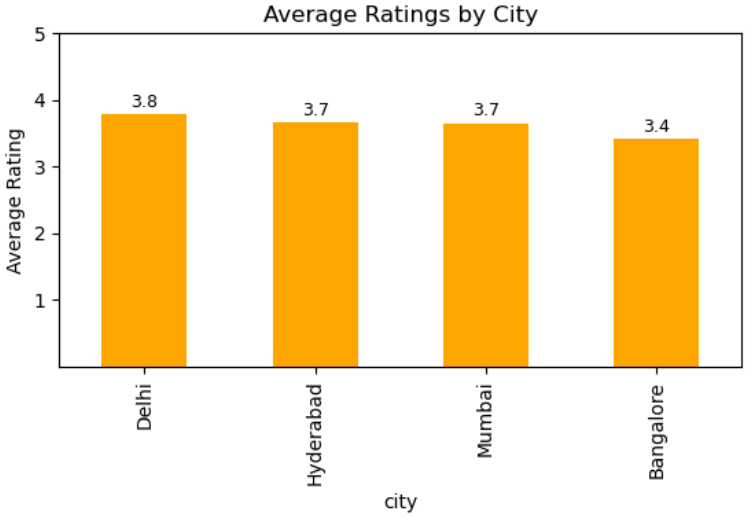
Exploring the distribution of revenue_realized reveals a classic right-skewed pattern. Most bookings cluster around moderate revenue ranges, while a smaller number of high-value bookings stretch the distribution into a long tail.

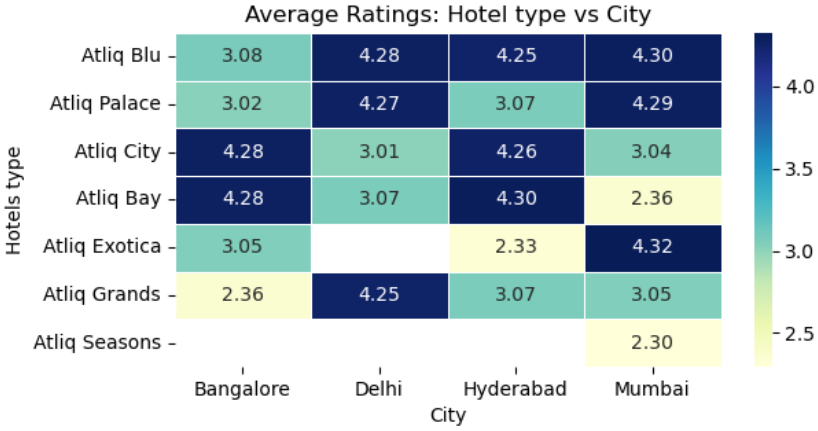
Even after removing extreme outliers, subtle peaks and troughs remain, reflecting natural groupings around common price points. This behavior aligns with typical booking patterns and reassures us that the dataset accurately represents real-world revenue.

2.RATING ANALYSIS









Diving into hotel ratings across room classes, cities, and properties uncovers clear patterns of guest satisfaction. Presidential rooms consistently receive the highest ratings, averaging 3.69, while Premium rooms average slightly lower at 3.59, suggesting that top-tier accommodations are perceived favorably.

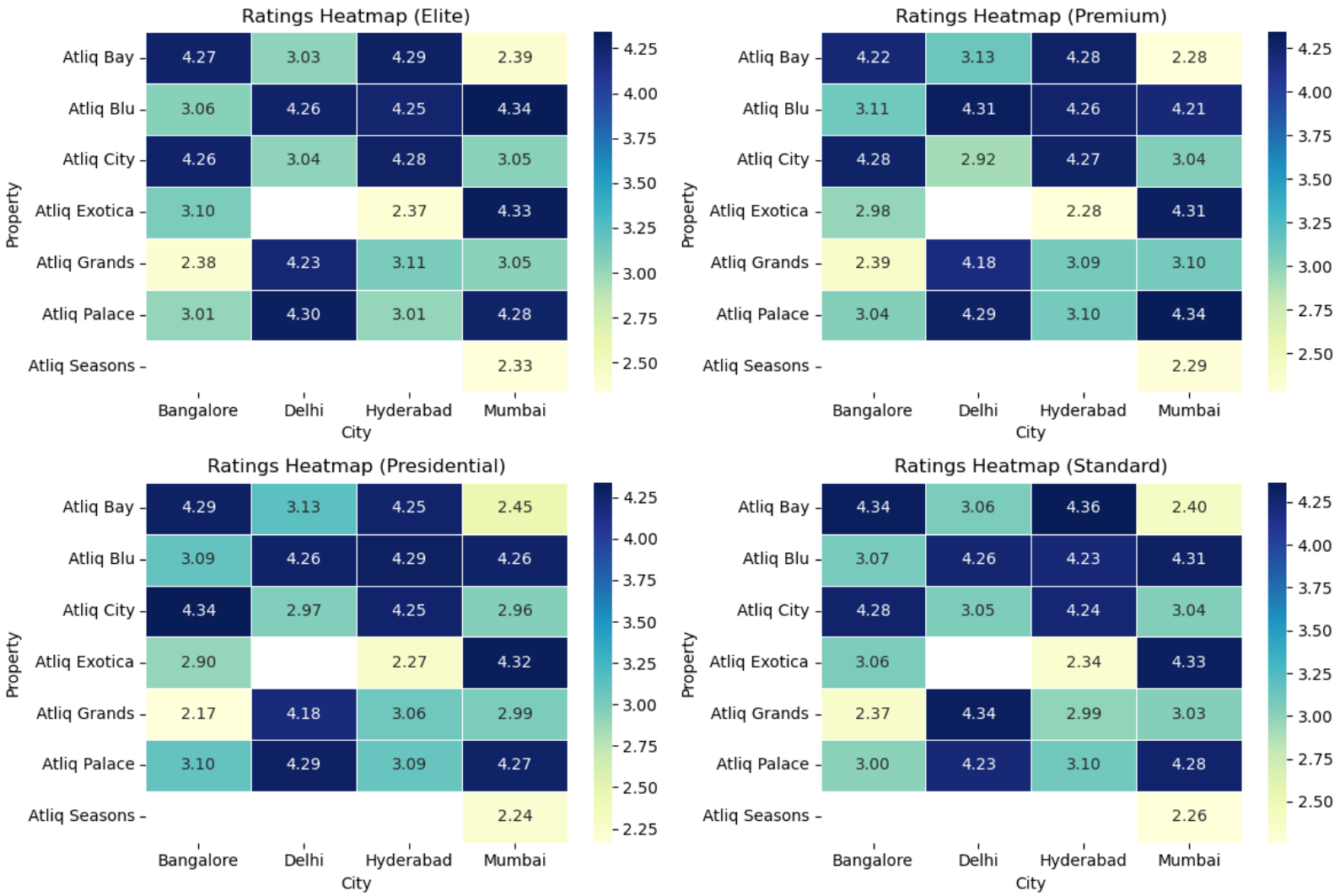
Across cities, Delhi stands out with the highest overall satisfaction at 3.78, while Bangalore trails at 3.41, signaling potential areas for improvement. Hyderabad and Mumbai fall in between, demonstrating moderate guest satisfaction.

At the hotel level, performance varies further. Atliq Blu impresses with a rating of 3.96, whereas Atliq Seasons lags at 2.30.

Location also plays a role: Atliq Exotica scores 4.32 in Mumbai but drops sharply to 2.33 in Hyderabad, while similar variations appear for Atliq Bay and Atliq Blu across cities.

These trends indicate that guest satisfaction depends not only on hotel type but also on city-specific factors such as management, service quality, and local expectations.

3. LOW-RATING HOTEL TYPES AND URGENT FOCUS



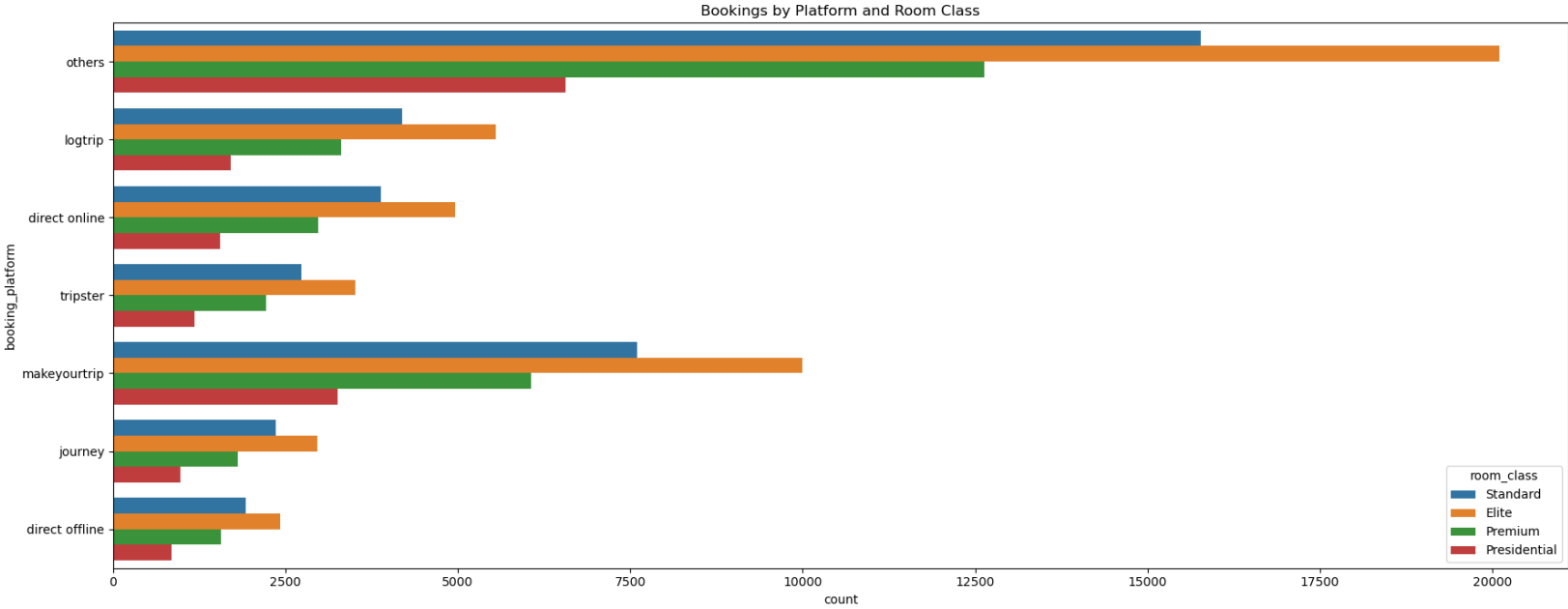
Examining low-performing properties, certain hotel-type, room-class, and city combinations emerge as urgent focus areas.

Atliq Seasons in Mumbai struggles across all room classes, while Atliq Exotica in Hyderabad underperforms in Presidential, Elite, Premium, and Standard rooms. Atliq Grands in Bangalore and Atliq Bay in Mumbai also show concerning ratings in multiple room classes.

These patterns suggest systemic challenges likely tied to service quality, amenities, or location-specific factors.

Addressing these issues will require a careful investigation of guest feedback, operational processes, and facilities to prioritize interventions and improve service delivery.

4. BOOKING INSIGHTS: ROOM CLASS AND PLATFORM



Guest preferences reveal a consistent story. Elite rooms dominate bookings across all major cities, reflecting strong demand for premium experiences. Standard and Premium rooms attract moderate interest, while Presidential rooms remain rare, likely reserved for special occasions or corporate stays.

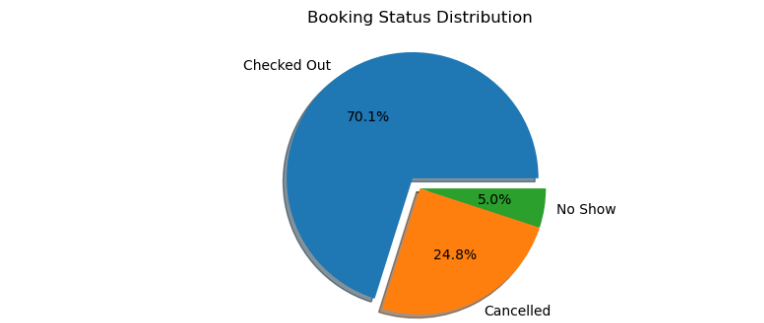
Booking platforms highlight additional patterns. The “others” platform accounts for the majority of reservations, followed by makeyourtrip and logtrip.

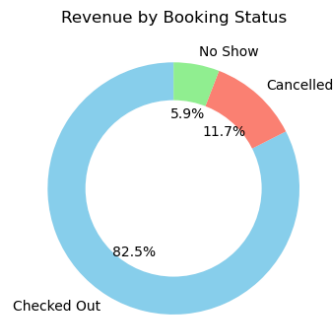
Across platforms, Elite rooms consistently lead, indicating high-end preferences are stable. Standard and Premium rooms maintain steady bookings, while Presidential rooms continue to see limited adoption.

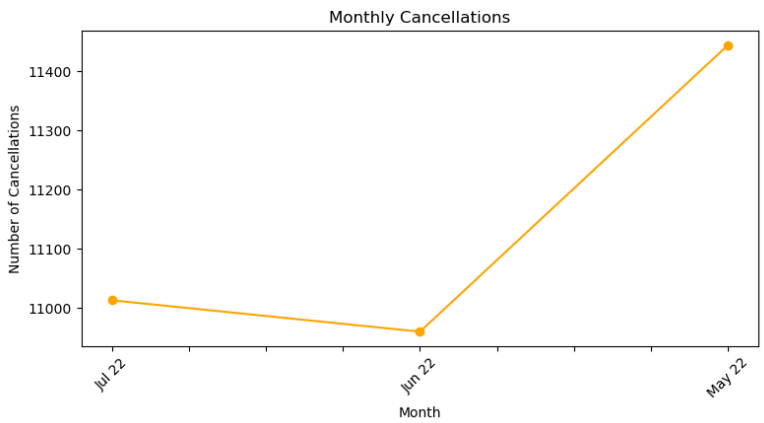
City-level booking counts reinforce this trend: Mumbai leads with 43,454 bookings, followed by Hyderabad (34,883), Bangalore (32,015), and Delhi (24,226). This shows the core guest base is concentrated in Mumbai.

Overall, Elite rooms form the backbone of revenue, while opportunities exist to grow demand for other room types, particularly on underperforming platforms, through targeted promotions or special offers.

5. BOOKING STATUS AND CANCELLATION ANALYSIS







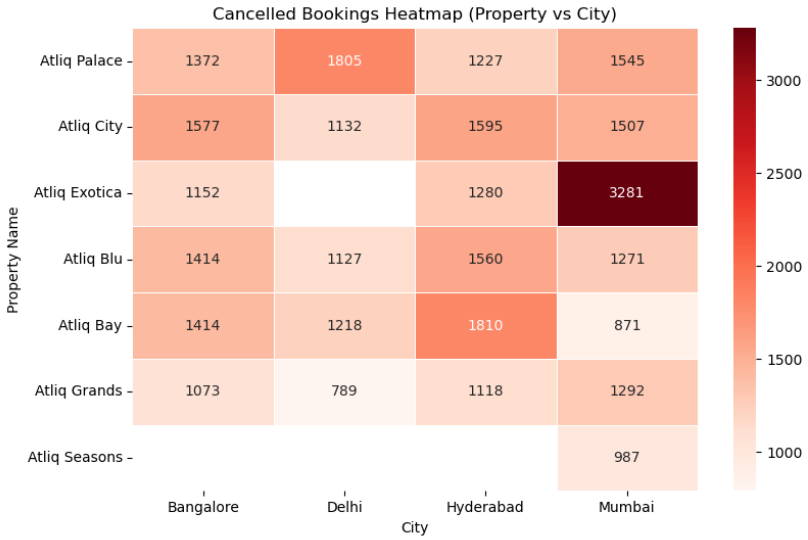
Analyzing bookings by status highlights the foundation of our revenue. Checked Out bookings dominate with 94,403 completed stays, compared to 33,417 cancellations and 6,758 no-shows.

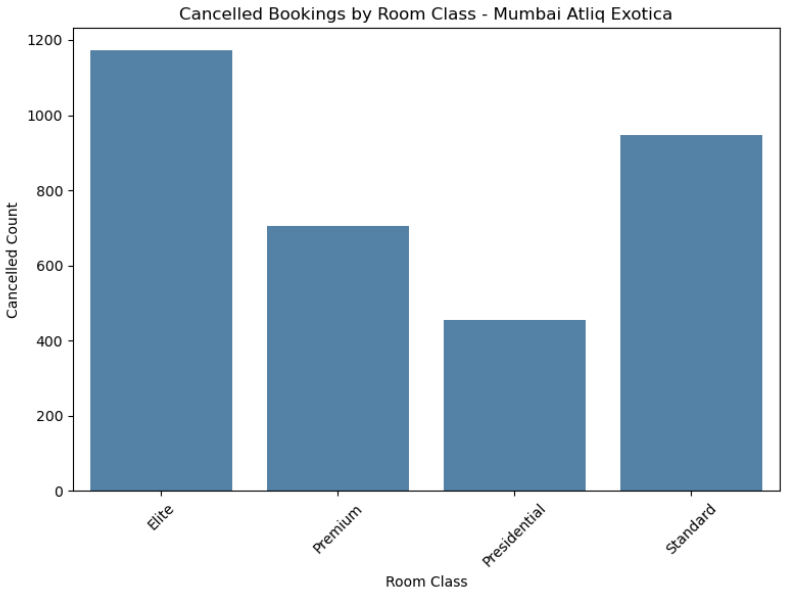
Revenue mirrors this pattern: Checked Out bookings generate 1,408,980,515, far exceeding Cancelled (199,170,324) and No Show (100,453,040) revenue.

This confirms that completed stays are the primary revenue drivers, while cancellations and no-shows represent lost opportunities.

Examining cancellations by city, Mumbai leads with 10,754, followed by Hyderabad (8,590), Bangalore (8,002), and Delhi (6,071).

Monthly cancellations are relatively consistent, with May at 11,444, June at 10,960, and July at 11,013, suggesting that cancellations are spread across months rather than concentrated.

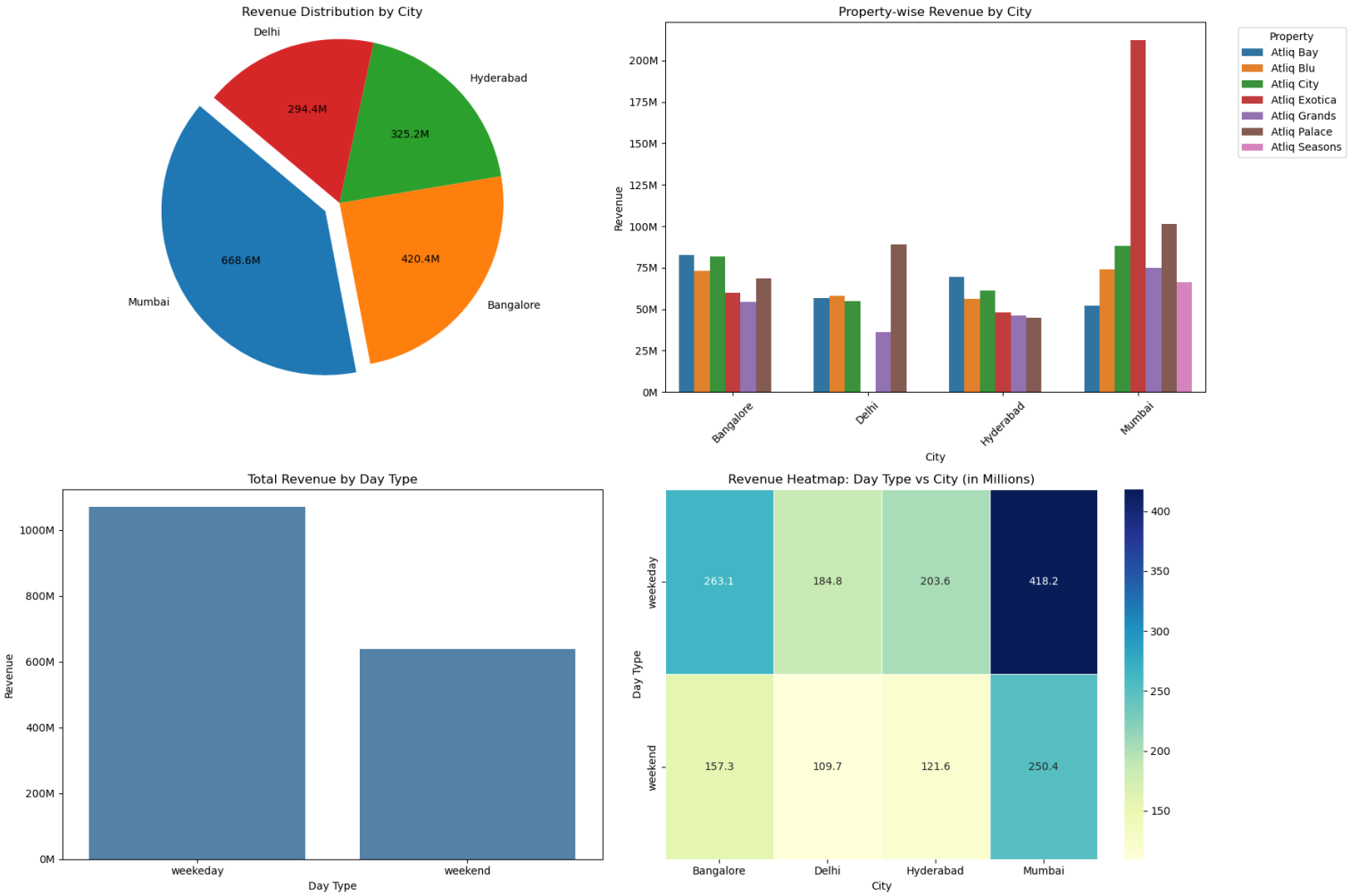




Drilling deeper, cancellations are not evenly distributed across hotels. In Mumbai, Atliq Exotica stands out with 1,173 cancelled Elite room bookings, followed by Standard (948), Premium (704), and Presidential (456).

These high-cancellation room classes represent the largest potential revenue loss and highlight where targeted interventions—flexible booking policies, reminders, or promotional incentives—could stabilize bookings and protect revenue.

6. REVENUE ANALYSIS SUMMARY

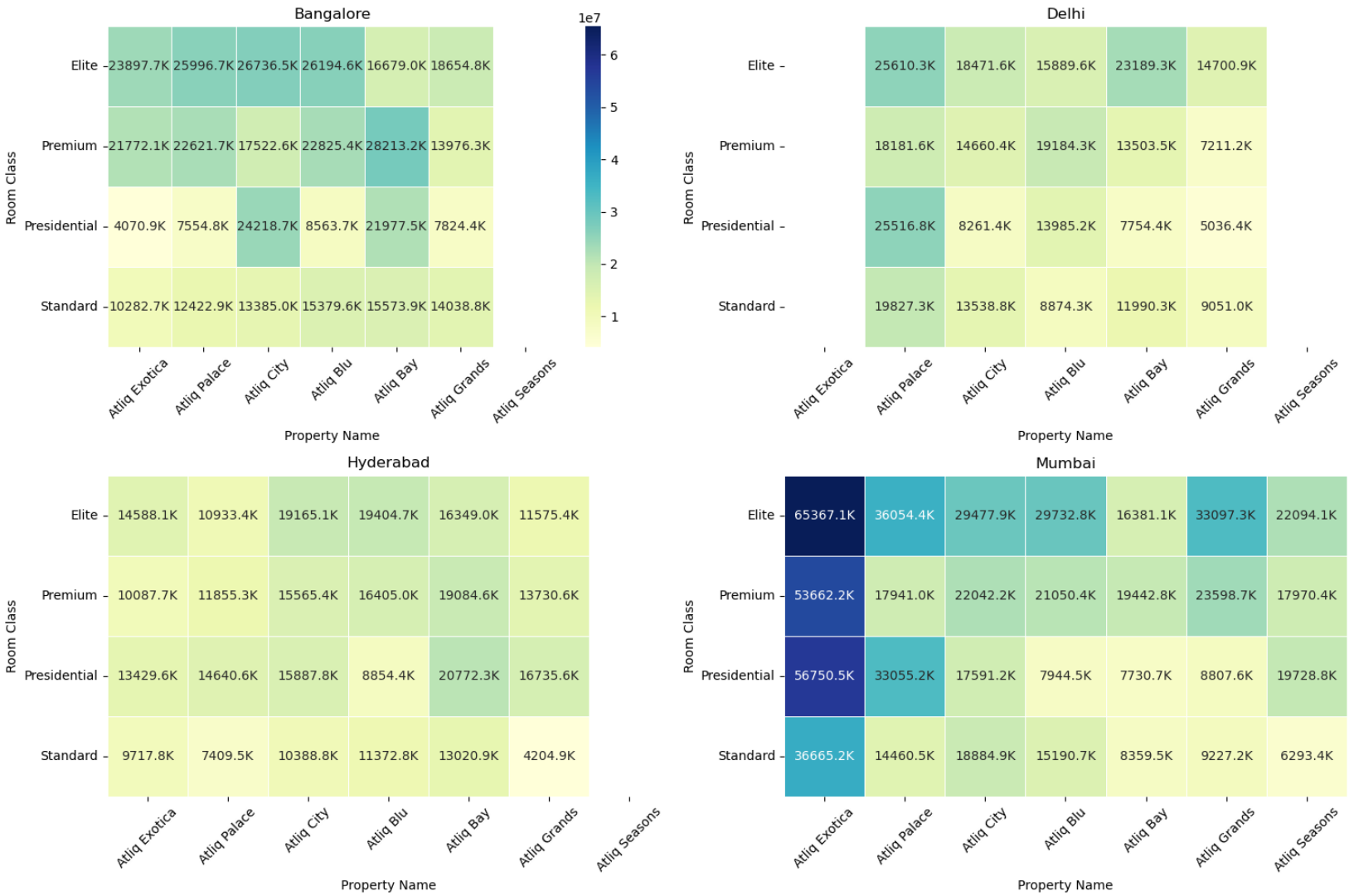


Weekdays consistently generate the bulk of revenue, a trend observed across all cities.

At the city level, revenue follows a clear hierarchy: Mumbai leads with 668.6M, followed by Bangalore (420.4M), Hyderabad (325.2M), and Delhi (294.4M).

While these figures might suggest underperformance in some cities, a deeper look highlights the impact of portfolio composition. Mumbai's strong revenue is driven by a concentration of high-performing hotel types, whereas Hyderabad, Delhi, and Bangalore show gaps in certain hotel types or room categories.

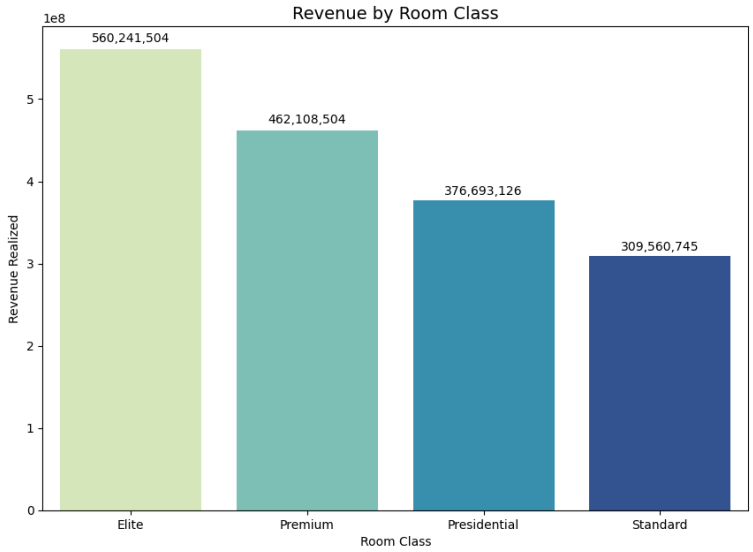
In other words, a city may seem to underperform not due to low demand, but because the optimal hotel types are not present a factor explored further in subsequent analysis.

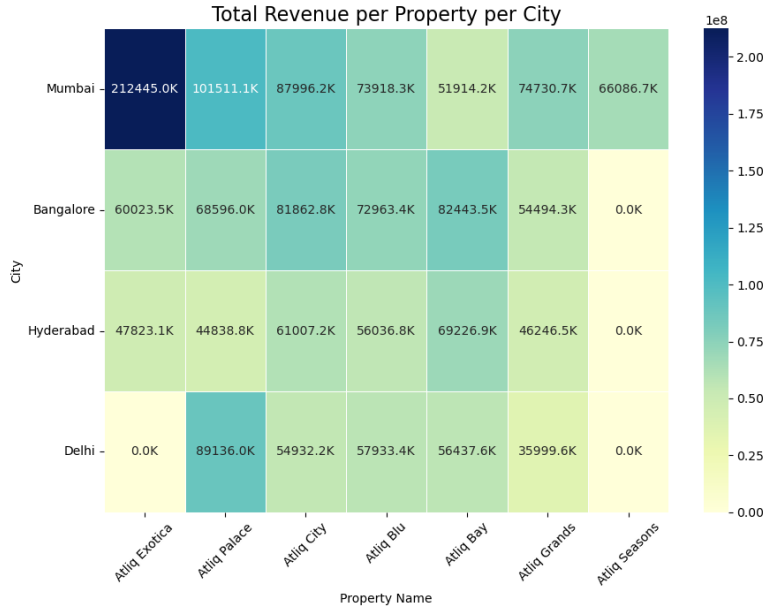


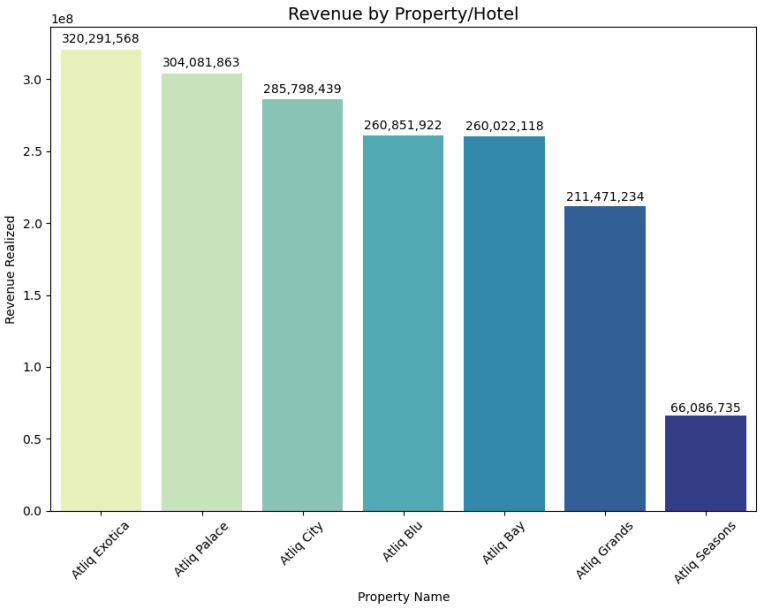
Analyzing revenue by hotel type and room class across cities provides a more granular view of performance. Some hotels consistently drive higher revenue, while certain room classes underperform in specific cities.

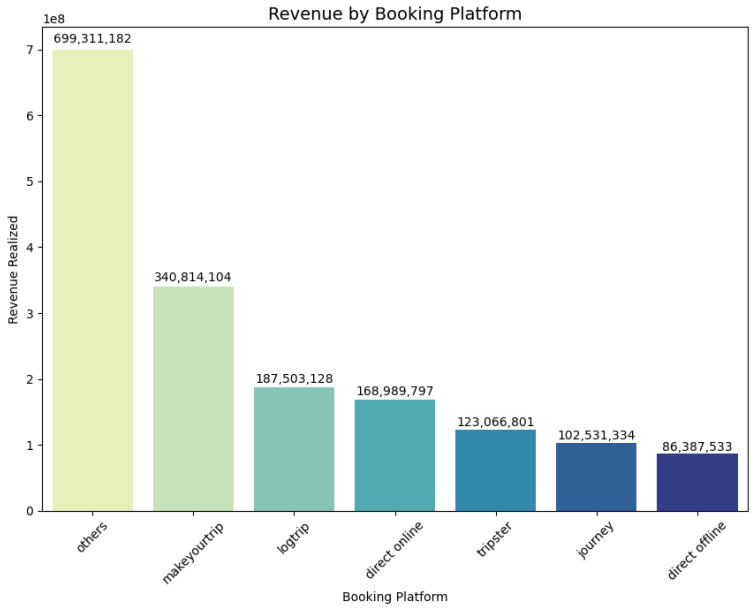
Examining revenue at both the city and property levels enables us to identify gaps and prioritize strategies for portfolio expansion, rebalancing, and targeted promotions to maximize overall revenue.

7.REVENUE OVERVIEW BY ROOM, PROPERTY, AND PLATFORM







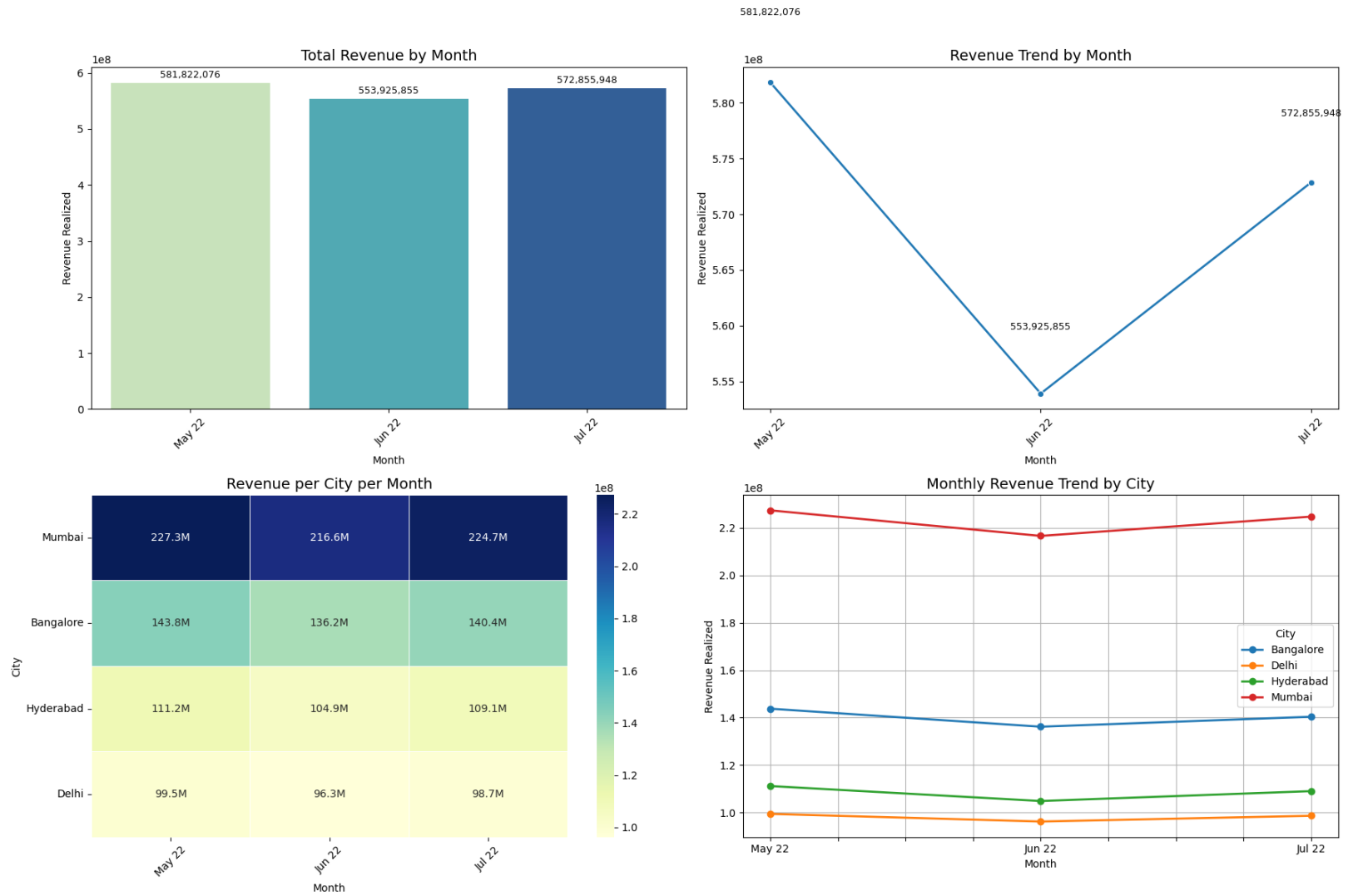


Exploring revenue by room class, property, and booking platform reveals a clear story.

Elite rooms drive the majority of revenue, forming the backbone of the business. Property-level heatmaps show which hotels contribute most in each city, while revenue by hotel type highlights consistently high-performing properties.

Revenue by booking platform indicates where marketing and distribution efforts are most effective. Together, these insights provide a holistic view of revenue concentration, performance drivers, and opportunities for growth.

8. MONTHLY REVENUE INSIGHTS

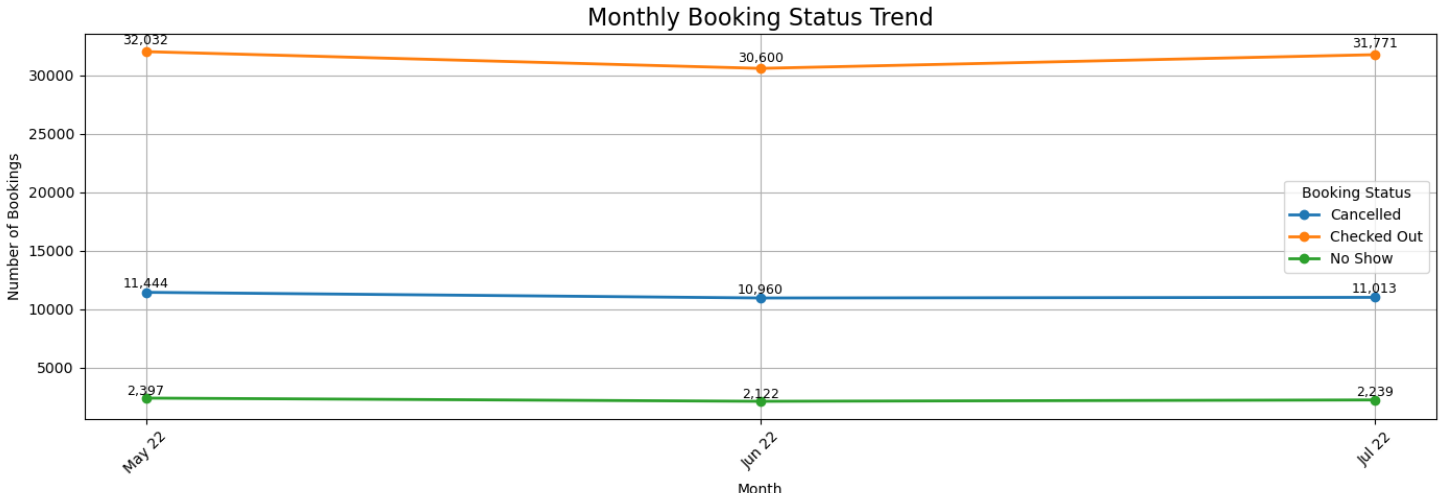


Month-by-month revenue trends uncover interesting patterns. May 2022 peaked at 581.8M, dropped in June to 553.9M, and rebounded in July to 572.9M.

The June dip is consistent across all cities—Bangalore, Delhi, Hyderabad, and Mumbai indicating that it is not tied to a specific market but reflects a broader trend affecting bookings.

Monitoring these fluctuations helps identify periods that may require targeted strategies to stabilize revenue.

9.POSSIBLE CAUSE OF JUNE REVENUE DIP

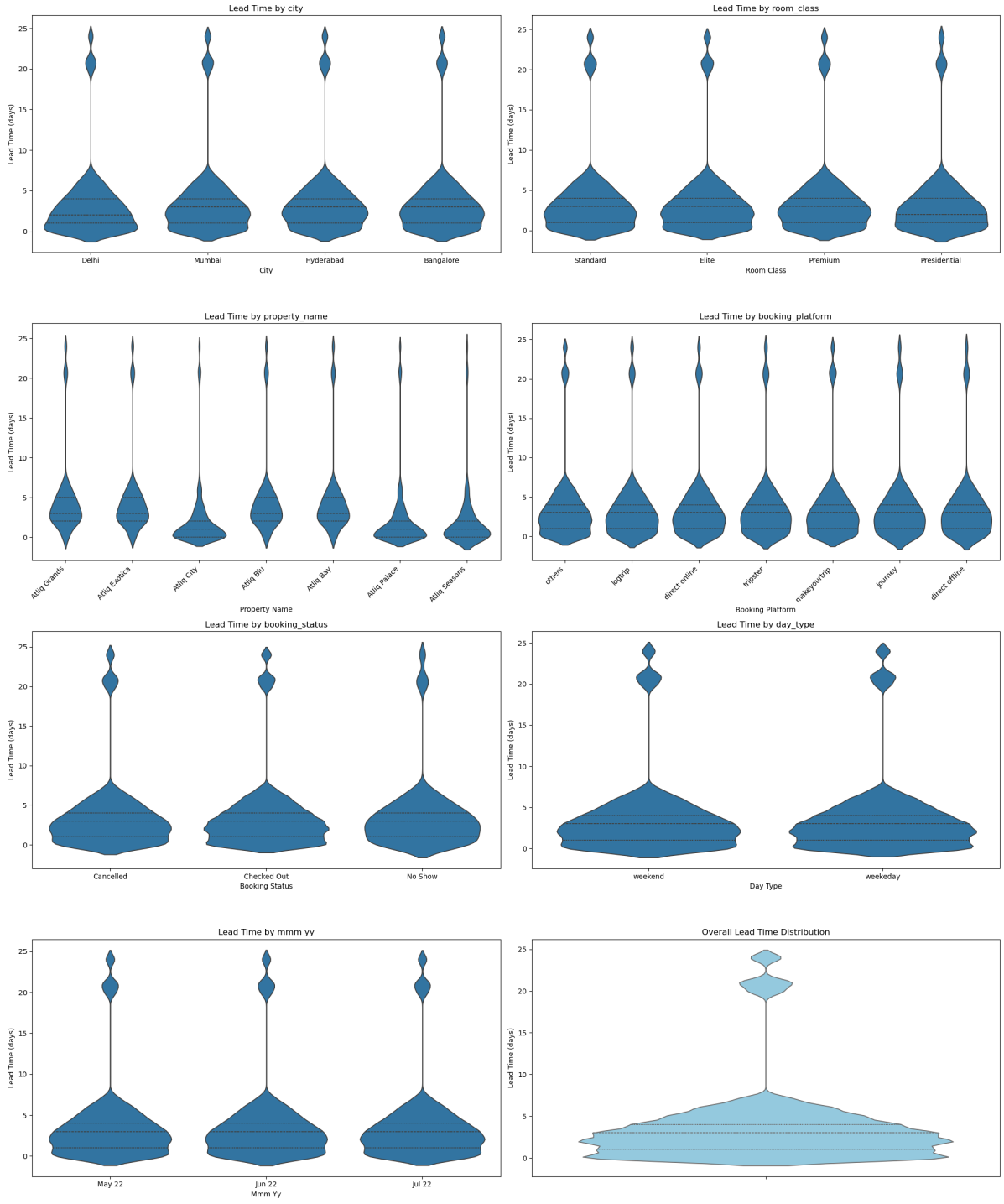


Closer inspection shows that the June dip is tied to a slight decline in completed stays. Checked Out bookings fell from 32,032 in May to 30,600 in June.

Cancellations and no-shows remained stable, confirming that the drop in revenue is operational rather than market-driven.

Understanding this helps clarify that revenue fluctuations are linked to the number of completed stays rather than a change in guest demand.

10.BOOKING LEAD TIME



Across cities, room classes, booking platforms, and months, lead time remains stable at 3–4 days.

This consistency indicates that changes in guest planning behavior are unlikely to drive revenue fluctuations

Note: These findings are preliminary and may be refined or expanded with further analysis if needed